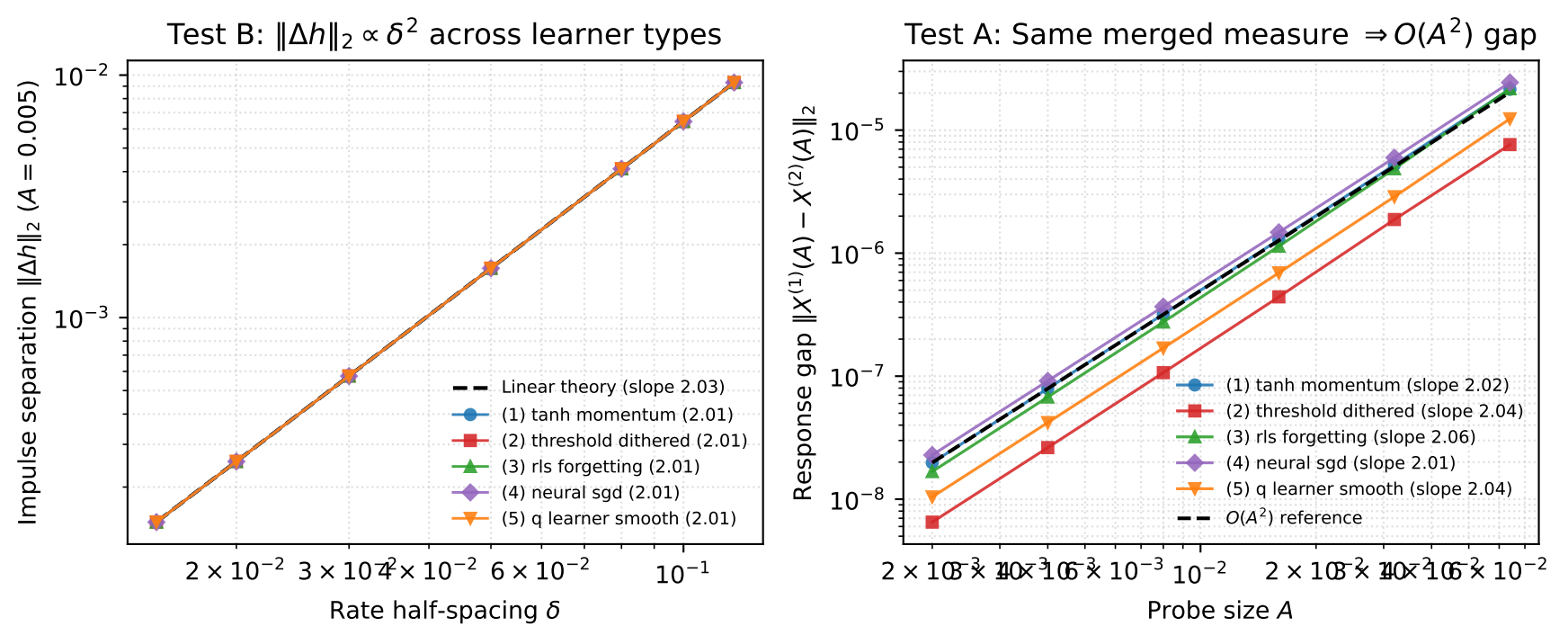
Test B: $\|\Delta h\|_2 \propto \delta^2$ across learner types
Impulse separation $\|\Delta h\|_2$ ($A = 0.005$)
$10^{-2}$
$10^{-3}$
$2 \times 10^{-2}$
$3 \times 10^{-2}$
$4 \times 10^{-2}$
$6 \times 10^{-2}$
$10^{-1}$
Rate half-spacing $\delta$
Linear theory (slope 2.03)
(1) tanh momentum (2.01)
(2) threshold dithered (2.01)
(3) rls forgetting (2.01)
(4) neural sgd (2.01)
(5) q learner smooth (2.01)
Test A: Same merged measure $\Rightarrow O(A^2)$ gap
Response gap $\|X^{(1)}(A) - X^{(2)}(A)\|_2$
$10^{-5}$
$10^{-6}$
$10^{-7}$
$10^{-8}$
$10^{-2}$
$2 \times 10^{-2}$
Probe size $A$
(1) tanh momentum (slope 2.02)
(2) threshold dithered (slope 2.04)
(3) rls forgetting (slope 2.06)
(4) neural sgd (slope 2.01)
(5) q learner smooth (slope 2.04)
$O(A^2)$ reference